Diagonalization Reading

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If D is a diagonal matrix, then it is relatively easy to analyze the behavior of the linear transformation $T(\mathbf{x}) = D\mathbf{x}$ because for $\mathbf{x}$ in $\mathbf{R}^n$,

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{bmatrix} \Rightarrow D\mathbf{x} = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} d_{11}x_1 \\ d_{22}x_2 \\ \vdots \\ d_{nn}x_n \end{bmatrix}$$

In this lesson we develop a procedure for expressing a square matrix A as the product of three matrices. The process is called *diagonalizing A*, because the middle matrix in the product is diagonal.

DEFINITION 8

An $n \times n$ matrix A is diagonalizable if there exist $n \times n$ matrices D and P, with D diagonal and P invertible, such that

$$A = PDP^{-1}$$

Because D is a diagonal matrix, expressing $A = PDP^{-1}$ makes it easier to analyze the linear transformation $T(\mathbf{x}) = A\mathbf{x}$. Diagonalizing A also allows for more efficient computation of matrix powers $A^2, A^3, \ldots$, which arise in modeling systems that evolve over time. Matrix powers are discussed at the end of the lesson.

Example 1

Let $A = \begin{bmatrix} -2 & 2 \\ -6 & 5 \end{bmatrix}$. Show that if

$$P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

then $A = PDP^{-1}$ and hence A is diagonalizable.

Solution

Applying the Quick Formula for the inverse of a 2×2 matrix gives us $P^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$. Thus we have

$$PDP^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 2 \\ -6 & 5 \end{bmatrix} = A$$

Therefore A is diagonalizable.

Now let's develop a method for finding matrices P and D that diagonalize an $n \times n$ matrix A. Suppose that A has n linearly independent eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Define

$$P = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}$$

Since $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are eigenvectors of A, we have

$$A\mathbf{u}_i = \lambda_i \mathbf{u}_i \quad \text{for } i = 1, \ldots, n$$

Hence it follows that

$$\begin{aligned} AP &= A \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix} \\ &= \begin{bmatrix} A\mathbf{u}_1 & A\mathbf{u}_2 & \cdots & A\mathbf{u}_n \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 \mathbf{u}_1 & \lambda_2 \mathbf{u}_2 & \cdots & \lambda_n \mathbf{u}_n \end{bmatrix} \end{aligned}$$

We also have

$$\begin{aligned} PD &= \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 \mathbf{u}_1 & \lambda_2 \mathbf{u}_2 & \cdots & \lambda_n \mathbf{u}_n \end{bmatrix} \end{aligned}$$

Thus $AP = PD$. Since $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly independent, P is an invertible matrix so that

$$A = A(PP^{-1}) = (AP)P^{-1} = (PD)P^{-1} = PDP^{-1}$$

and A is diagonalizable.

Example 2

Find matrices P and D to diagonalize $A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}$.

Solution

We start by finding the eigenvalues and eigenvectors of A. Beginning with the eigenvalues, we have

$$\begin{aligned} \det(A - \lambda I_2) &= \begin{vmatrix} 3 - \lambda & 1 \\ -2 & -\lambda \end{vmatrix} \\ &= (3 - \lambda)(-\lambda) - (-2) = \lambda^2 - 3\lambda + 2 = (\lambda - 2)(\lambda - 1) \end{aligned}$$

Thus we have eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$. Starting with $\lambda_1 = 1$, the homogeneous system $(A - \lambda_1 I)\mathbf{u} = \mathbf{0}$ has augmented matrix and echelon form

$$\begin{bmatrix} 2 & 1 & 0 \\ -2 & -1 & 0 \end{bmatrix} \quad \begin{matrix} R_1 + R_2 \to R_2 \\ \sim \end{matrix} \quad \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Back substitution gives us the associated eigenvector

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

For $\lambda_2 = 2$, the system $(A - 2I)\mathbf{u} = \mathbf{0}$ has augmented matrix and echelon form

$$\begin{bmatrix} 1 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \quad \begin{matrix} 2R_1 + R_2 \to R_2 \\ \sim \end{matrix} \quad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This time back substitution gives us the eigenvector

$$\mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Now we define P and D, with

$$P = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

To check our work, we find P^{-1} using the Quick Formula, and then compute the product

$$PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix} = A$$

We can use the diagonalization formula to construct a matrix A that has specified eigenvalues and eigenvectors.

Example 3

Find a 2×2 matrix A that has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ and corresponding eigenvectors $\mathbf{u}_1 = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Solution

We start by defining the diagonalization matrices P and D and multiply to find A. Let

$$P = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$$

We have $P^{-1} = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}$ from the Quick Formula for 2×2 matrix inverses, so that

$$A = PDP^{-1} = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix} = \begin{bmatrix} -28 & 45 \\ -18 & 29 \end{bmatrix}$$

To verify that A has the required specifications, we compute

$$\begin{aligned} A\mathbf{u}_1 &= \begin{bmatrix} -28 & 45 \\ -18 & 29 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \end{bmatrix} = (-1) \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \lambda_1 \mathbf{u}_1 \\ A\mathbf{u}_2 &= \begin{bmatrix} -28 & 45 \\ -18 & 29 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = (2) \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \lambda_2 \mathbf{u}_2 \end{aligned}$$

Hence $\{\lambda_1, \lambda_2\}$ are eigenvalues of A associated with eigenvectors $\{\mathbf{u}_1, \mathbf{u}_2\}$.

Many square matrices are diagonalizable, but not all. The next theorem tells us exactly when a matrix is diagonalizable.

THEOREM 9

An $n \times n$ matrix A is diagonalizable if and only if A has eigenvectors that form a basis for $\mathbf{R}^n$.

Proof

We have seen how to diagonalize A if A has eigenvectors that form a basis for $\mathbf{R}^n$, so half of the proof is done. For the other half, suppose that A is diagonalizable, with

$$A = PDP^{-1} \quad (1)$$

where $\mathbf{p}_1, \ldots, \mathbf{p}_n$ are the columns of P and $d_{11}, \ldots, d_{nn}$ are the diagonal entries of D. Since P is invertible, the columns $\mathbf{p}_1, \ldots, \mathbf{p}_n$ of P are nonzero and linearly independent.

Multiplying by P on the right of both sides of (1), we have $AP = PD$. Since column i of AP is equal to $A\mathbf{p}_i$ and column i of PD is equal to $d_i\mathbf{p}_i$, we have

$$A\mathbf{p}_i = d_{ii}\mathbf{p}_i$$

Therefore $\mathbf{p}_i$ is an eigenvector of A with associated eigenvalue d_{ii}. Since P is invertible, A has eigenvectors that form a basis for $\mathbf{R}^n$.

The proof tells us both when A is diagonalizable and how A is diagonalized. The diagonal elements of D must be the eigenvalues and the columns of P must be associated eigenvectors. Of course, there are many possibilities for the associated eigenvectors, so the diagonalization is not unique. However, we now have a way to find a diagonalization of an $n \times n$ matrix A.

Summary: Diagonalizing an $n \times n$ matrix A

Find the eigenvalues and the associated linearly independent eigenvectors.

- If A has n linearly independent eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$, then A is diagonalizable, with $P = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix}$ and the diagonal entries of D given by the corresponding eigenvalues.
- If there are not n linearly independent eigenvectors, then A is not diagonalizable.

Note that the order of the eigenvalues in D does not matter, as long as it matches the order of the corresponding eigenvectors in P.

The next theorem tells us that eigenvectors associated with distinct eigenvalues must be linearly independent. This theorem comes in handy when trying to diagonalize a matrix.

THEOREM 10

If $\{\lambda_1, \ldots, \lambda_r\}$ are distinct eigenvalues of a matrix A, then any set of associated eigenvectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is linearly independent.

Proof

Suppose that the set of eigenvectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is linearly dependent. Since all eigenvectors are nonzero, one of the eigenvectors can be written as a linear combination of a linearly independent subset of the remaining eigenvectors, with the coefficients nonzero and unique for the given subset. Thus, without loss of generality, let $c_1, \ldots, c_j$ be nonzero scalars such that

$$\mathbf{u}_1 = c_2\mathbf{u}_1 + \cdots + c_j\mathbf{u}_j \quad (2)$$

Then

$$\begin{aligned} \lambda_1\mathbf{u}_1 &= A\mathbf{u}_1 = A(c_2\mathbf{u}_2 + \cdots + c_j\mathbf{u}_j) \\ &= c_2A\mathbf{u}_2 + \cdots + c_jA\mathbf{u}_j \\ &= c_2\lambda_2\mathbf{u}_2 + \cdots + c_j\lambda_j\mathbf{u}_j \end{aligned}$$

If $\lambda_1 \neq 0$, then

$$\mathbf{u}_1 = c_2\left(\frac{\lambda_2}{\lambda_1}\right)\mathbf{u}_2 + \cdots + c_j\left(\frac{\lambda_j}{\lambda_1}\right)\mathbf{u}_j$$

which is a different linear combination equal to $\mathbf{u}_1$, contradicting the uniqueness of (2). If $\lambda_1 = 0$, then

$$c_2\lambda_2\mathbf{u}_2 + \cdots + c_j\lambda_j\mathbf{u}_j = \mathbf{0}$$

Since $c_2, \ldots, c_j$ and $\lambda_2, \ldots, \lambda_j$ are nonzero, this contradicts the linear independence of $\{\mathbf{u}_2, \ldots, \mathbf{u}_j\}$. Hence either way we reach a contradiction, so it must be that the set of eigenvectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is linearly independent.

Example 4

If possible, diagonalize the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$.

Solution

We start by finding the eigenvalues by factoring the characteristic polynomial of A,

$$\det(A - \lambda I_3) = -\lambda^3 - 2\lambda^2 - \lambda = -\lambda(\lambda + 1)^2$$

Thus we have eigenvalues $\lambda_1 = 0$ and $\lambda_2 = -1$. Starting with $\lambda_1 = 0$, the augmented matrix for the system $(A - 0I)\mathbf{u} = \mathbf{0}$ and the corresponding echelon form are

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -2 & -2 & -1 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \quad \begin{matrix} R_1 + R_2 \to R_2 \\ R_2 + R_3 \to R_3 \\ \sim \end{matrix} \quad \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Back substitution can be used to show that a basis for the eigenspace associated with $\lambda_1 = 0$ is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$.

For $\lambda_2 = -1$, the augmented matrix and echelon form are

$$\begin{bmatrix} 2 & 1 & 1 & 0 \\ -2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_1 + R_2 \to R_2 \\ \sim \end{matrix} \quad \begin{bmatrix} 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Back substituting shows that the eigenspace of $\lambda_2 = -1$ has dimension 2 and basis $\left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \right\}$. By Theorem 10 we know that eigenvectors associated with distinct eigenvalues are linearly independent. Hence the set

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \right\}$$

is linearly independent and thus forms a basis for $\mathbf{R}^3$. Since there are two linearly independent eigenvectors associated with $\lambda_2 = -1$, this eigenvalue appears twice in D. We have

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Note that the eigenvalues along the diagonal of D are in columns corresponding to the columns of P containing the associated eigenvectors. We check that D and P are correct by computing

$$PD = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

and

$$AP = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Since $AP = PD$ and P is invertible, we have $A = PDP^{-1}$.

The next theorem provides a set of conditions required for a matrix to be diagonalizable.

THEOREM 11

Suppose that an $n \times n$ matrix A has only real eigenvalues. Then A is diagonalizable if and only if the dimension of each eigenspace is equal to the multiplicity of the corresponding eigenvalue.

See Lesson 1, Eigenvalues and Eigenvectors, for the definition of the multiplicity of an eigenvector.

Proof

This theorem follows from things that we already know:

- An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. (This is from Theorem 9.)
- Each eigenspace has a dimension no greater than the multiplicity of its associated eigenvalue. (This is from Theorem 6 in Lesson 1.)
- If A is an $n \times n$ matrix, then the multiplicities of the eigenvalues sum to n. (This is because the degree of the characteristic polynomial is equal to n, and the multiplicities must add up to the degree.)
- Vectors from distinct eigenspaces are linearly independent. (This is from Theorem 10.)

Pulling these together, we see that A is diagonalizable when the dimension of each eigenspace is as large as possible. Otherwise there will not be enough linearly independent eigenvectors to form a basis. If the dimension of each eigenspace is as large as possible, then since vectors from distinct eigenspaces are linearly independent we know that there will be enough linearly independent eigenvectors to form a basis for $\mathbf{R}^n$.

Example 5

If possible, diagonalize the matrix $A = \begin{bmatrix} 3 & 6 & 5 \\ 3 & 2 & 3 \\ -5 & -6 & -7 \end{bmatrix}$.

Solution

The characteristic polynomial for A is

$$\det(A - \lambda I_3) = -(\lambda + 2)^2(\lambda - 2)$$

giving us eigenvalues $\lambda_1 = -2$ and $\lambda_2 = 2$. For $\lambda_1 = -2$, the augmented matrix for the system $(A + 2I)\mathbf{u} = \mathbf{0}$ and the corresponding echelon form are

$$\begin{bmatrix} 5 & 6 & 5 & 0 \\ 3 & 4 & 3 & 0 \\ -5 & -6 & -5 & 0 \end{bmatrix} \quad \begin{matrix} -\frac{3}{5}R_1 + R_2 \to R_2 \\ \frac{1}{5}R_1 + R_2 \to R_3 \\ \sim \end{matrix} \quad \begin{bmatrix} 5 & 6 & 5 & 0 \\ 0 & \frac{2}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Back substitution shows that the eigenspace associated with $\lambda_1 = -2$ has basis $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$.

This eigenspace has dimension 1, which is less than the multiplicity of $\lambda_1 = -2$. By Theorem 11, we know immediately that A is not diagonalizable.

We usually will need to find the eigenvectors in order to determine if a matrix is diagonalizable, but there is a case where knowing only the eigenvalues is enough.

THEOREM 12

If A is an $n \times n$ matrix with n distinct real eigenvalues, then A is diagonalizable.

Proof

Every eigenvalue has an eigenvector, ensuring that the associated eigenspace has dimension at least 1. On the other hand, if the eigenvalues are distinct, then each has multiplicity 1, so that each eigenspace must have dimension 1—there is no other option. Thus by Theorem 11 A is diagonalizable.

Example 6

If possible, diagonalize the matrix $A = \begin{bmatrix} 5 & 0 & 0 \\ -4 & 3 & 0 \\ 1 & -3 & -2 \end{bmatrix}$.

Solution

Since A is lower triangular, the eigenvalues lie along the main diagonal, with $\lambda_1 = 5$, $\lambda_2 = 3$, and $\lambda_3 = -2$. These are distinct, so we know from Theorem 12 that A is diagonalizable. Bases for the associated eigenspaces are (computations are not shown)

$$\lambda_1 = 5 \Rightarrow \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}, \quad \lambda_2 = 3 \Rightarrow \left\{ \begin{bmatrix} 0 \\ -5 \\ 3 \end{bmatrix} \right\}, \quad \lambda_3 = -2 \Rightarrow \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Therefore $A = PDP^{-1}$, with

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -5 & 0 \\ 1 & 3 & 1 \end{bmatrix}$$

Matrix Powers

Suppose that A is diagonalizable, with $A = PDP^{-1}$. Then

$$\begin{aligned} A^2 &= (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1} \\ A^3 &= A(A^2) = (PDP^{-1})(PD^2P^{-1}) = PD(P^{-1}P)D^2P^{-1} = PD^3P^{-1} \end{aligned}$$

and in general

$$A^k = PD^kP^{-1}$$

This formula for A^k can be formally proved using induction.

Next, note that

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix} \Rightarrow D^k = \begin{bmatrix} d_{11}^k & 0 & \cdots & 0 \\ 0 & d_{22}^k & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & d_{nn}^k \end{bmatrix}$$

For example, if $D = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$, then

$$\begin{aligned} D^2 &= \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} (-3)^2 & 0 \\ 0 & 2^2 \end{bmatrix} \\ D^3 &= \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} (-3)^2 & 0 \\ 0 & 2^2 \end{bmatrix} = \begin{bmatrix} (-3)^3 & 0 \\ 0 & 2^3 \end{bmatrix} \end{aligned}$$

and so on. Hence we see that although directly calculating A^k can take many computations, calculating D^k is relatively easy.

Example 7

Suppose that $A = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}$. Find P and D so that $A = PDP^{-1}$, and then use this to give a formula for A^k.

A is an example of a *probability matrix*, because the entries are nonnegative and each column adds to 1.

Solution

Leaving out the computational details, the eigenvalues and associated eigenvectors of A are

$$\lambda_1 = 1 \Rightarrow \mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \lambda_2 = \frac{1}{2} \Rightarrow \mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Therefore if

$$P = \begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

then $A = PDP^{-1}$. To compute A^k, we use

$$\begin{aligned} A^k = PD^kP^{-1} &= \begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & \left(\frac{1}{2}\right)^k \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 + 3\left(\frac{1}{2}\right)^k & 1 - \left(\frac{1}{2}\right)^k \\ 3 - 3\left(\frac{1}{2}\right)^k & 3 + \left(\frac{1}{2}\right)^k \end{bmatrix} \end{aligned}$$

Since $\left(\frac{1}{2}\right)^k \to 0$ as $k \to \infty$, it follows that $A^k \to \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{3}{4} \end{bmatrix}$ as $k \to \infty$.

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